# Accelerating cosmology

## Handley Lab: KICC Welcome Day

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# Accelerating cosmology from theory to data

**Handley Lab: Science@KICC** <wh260@cam.ac.uk>

- **Big problems**: cosmic acceleration – $\Lambda$, evolving dark energy or modified gravity? $H_0$, $S_8$ and DESI–SNe tensions: physics, systematics or priors?
- **GPUs for science**: classical Bayes, differential equations and forward models in `JAX` / `BlackJAX`.
- **AI for scientists**: agents for coding and mining legacy code for embodied intelligence. Scientists as pilots of this new scale of science.
- **The programmes**: dark energy and fundamental theory; real-time gravitational waves; LSST, Euclid and JWST readiness.



# The Handley Lab (2026–2027)



**Natalie Hogg**
Postdoc



**Matt Grayling**
Postdoc



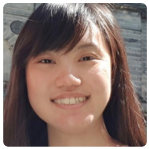
**Chris Lovell**
Postdoc



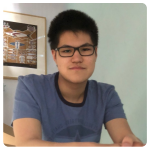
**James Alvey**
Kavli Senior Fellow



**Nikhil Sarin**
Kavli Senior Fellow



**Dily Ong**
PhD



**Namu Kroupa**
PhD




**Sam Leeney**
PhD



**Toby Lovick**
PhD



**Lawrence Berry**
PhD



**Charlotte Priestley**
PhD



**Farah Wallauer**
New PhD

SV

**Selma Vangstein**
New PhD

SH

**Sam Hewson**
New PhD

JL

**Jaiman Lad**
Part III

JGF

**Jesús Girones Fraile**
Part III

DP

**David Parodi**
Part III

MK

**Manaal Khan**
Part III

CM

**Charlotte McCarthy**
Part III

AB

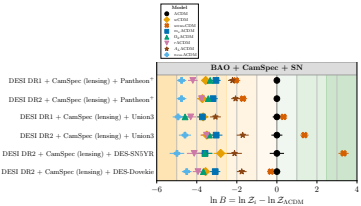
**Ashwin Balakrishnan**
Part III

BTB

**Benedict Thomas Binu**
Part III

# Dark energy, tensions and fundamental theory to data

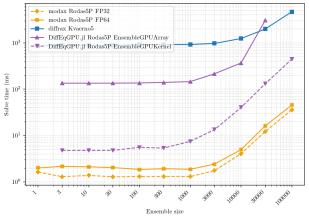
## DESI DR2, GPU Boltzmann solvers and candidate Lagrangians




**DESI: evolving dark energy?**
Dily Ong, David Yallup, Adam Ormondroyd

Separate evolving dark energy from priors and supernova calibration. [2603.05472]

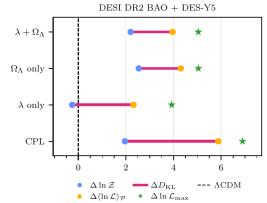
`unimpeded`: compare models and datasets using public chains. [2511.04661]




**GPU Boltzmann solvers**
Lawrence Berry, Oliver Hahn, Nils Schöneberg

`modax`: GPU differential equations.
`DISCO-EB`: CMB spectra without tight-coupling approximations. [2609.35661]

Next: evolve candidate theories and compare with data.




Sinah Legner: thesis, torsion models vs DESI

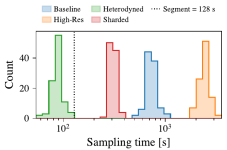
**Fundamental actions to data**
Sinah Legner, Wei-Ning Deng, Selma Vangstein, Sam Hewson

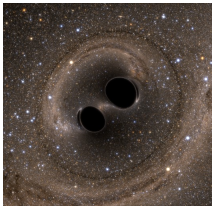
Torsion Condensation and CPT-symmetric cosmology. [2507.09228] [2509.10379]

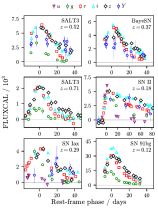
Next: agent-derived pipeline from Lagrangian to data, with full perturbations.

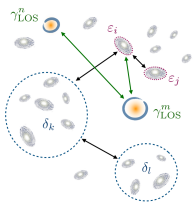
# Real-time GWs and Stage-IV survey readiness

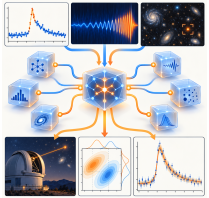
## Classical Bayes for LVK, Einstein Telescope, LISA, LSST, Euclid and JWST





**Real-time gravitational waves**
Metha Prathaban, David Yallup, James Alvey, Nikhil Sarin, Farah Wallauer

GPU likelihoods and `SwiG` nested sampling for binary neutron-star inference. [2607.28265]

Next: Einstein Telescope, LISA, overlapping signals and shadow sirens.

Black-hole merger: SXS simulation / LIGO

**Time-domain surveys and lensing**
Toby Lovick, Matt Grayling, Namu Kroupa, Sam Leeney, Charlotte Priestley; Natalie Hogg, Charlotte McCarthy, Chris Lovell

ALCS: marginalise latent structure. [2603.26644] `JAX-SNANA`: simulate supernovae. [2610.19000] Quasar delays [2602.22307]; radio transients and satellite orbits.

Euclid / JWST lensing; CAMELS / FLAMELS simulations.

**Agentic decision making at scale**
James Alvey, Nikhil Sarin

Next: transient tagging, model selection and re-fitting.

Agents: generated illustration

# The Ambient AI Lab: AI for scientists, by scientists

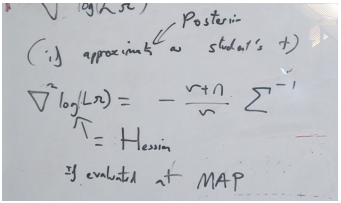
## Cambridge-owned compute, live room loops and how to join in



**Sovereign compute and tools**

Local models and transcription on `turing` / `newton`; Cambridge UIS compute.

`Alan` / `Steve`: agent tools using our shared scientific context.

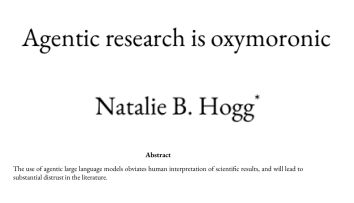



**Ambient and multiplayer AI**

Toby Lovick, James Alvey, Chris Lovell, Natalie Hogg

Whiteboard capture, local transcription, equation retrieval and shared supervision context.

**This term**: `ai@cam` supervision hardware; Cosmos ambient capture and anti-sycophancy projects.


Agentic research is oxymoronic

Natalie B. Hogg*

Abstract

The use of agentic large language models obviates human interpretation of scientific results, and will lead to substantial distrust in the literature.

**Training without deskilling**

Natalie Hogg: Nature Astronomy [2608.31161]

Students build physical intuition; senior researchers supervise agents.

People select questions, test the physics and take responsibility.

**Join us**: Tuesday group meeting, 14:00; Friday arXiv Scrolling Club; Friday Cavendish Gravity, 15:00 (F22); weekly Transients & GWs. `handley-lab.co.uk`